Q1 - 2024 (05 Apr Shift 2)

A sonometer wire of resonating length 90 cm has a fundamental frequency of 400 Hz when kept under some tension. The resonating length of the wire with fundamental frequency of 600 Hz under same tension _____cm .

Q2 - 2024 (06 Apr Shift 2)

Two open organ pipes of lengths 60 cm and 90 cm resonate at 6^{th} and 5^{th} harmonics respectively. The difference of frequencies for the given modes is _______ Hz. (Velocity of sound in air $= 333 \mathrm{~m/s}$)

Q3 - 2024 (08 Apr Shift 1)

A closed and an open organ pipe have same lengths. If the ratio of frequencies of their seventh overtones is $\left(\frac{a-1}{a}\right)$ then the value of a is ________

Q4 - 2024 (08 Apr Shift 2)

A plane progressive wave is given by $y = 2 \cos 2\pi(330\mathrm{t} - x)\mathrm{m}$. The frequency of the wave is :

(1) 330 Hz

(2) 660 Hz

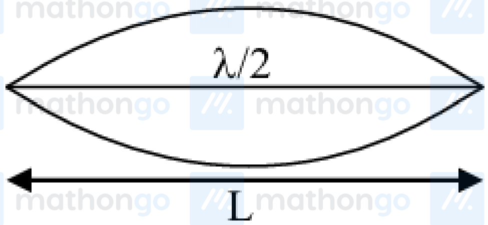
(3) 340 Hz

(4) 165 Hz

Answer Key

Q1 (60) **Q2** (740) **Q3** (16) **Q4** (1)

Q1



$f_0 = 400\text{ Hz}; v = \sqrt{\frac{T}{\mu}} = \text{ constant}$

$\frac{\lambda}{2} = L; \quad v = f_0\lambda$

$\frac{v}{2f_0} = L \Rightarrow v = 2Lf_0$

$L' = \frac{v}{2f'} = \frac{2Lf_0}{2f'}$

$= \frac{Lf_0}{f'} = \frac{90 \times 400}{600} = 60$

Q2

The difference in frequency in open organ pipe =

$f = \frac{nv}{2\,L}$

$\Delta f = \frac{6v}{2 \times 0.6} - \frac{5v}{2 \times 0.9}$

$v = 333\text{ m/s}$

$\Delta f = 740\text{ Hz}$

Q3

For closed organ pipe

$f_c = (2n+1)\frac{v}{4\ell} = \frac{15v}{4\ell}$

For open organ pipe

$f_o = (n+1)\frac{v}{2\ell} = \frac{8v}{2\ell}$

$\frac{f_c}{f_o} = \frac{15}{16} = \frac{a-1}{a}$

$\Rightarrow a = 16$

Q4

$y = 2\cos 2\pi(330t - x)m$
$y = A\cos(\omega t - kx)$
by comparing $\omega = 2\pi \times 330$
$2\pi f = 2\pi \times 330$
$f = 330$